

Online Evaluation

Information Retrieval

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Evaluation 101

- Suppose, we've implemented a change in a search engine:
 - New learning to rank method
 - New [neural] ranking feature
 - Change in the user interface
 - ...

Evaluation problem: will it improve the users' experience? Is it worth deploying at all?

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Evaluation Methods

- Three families of evaluation methods widely used both in the literature and in practice
 - Offline evaluation
 - User study evaluation
 - Online evaluation
- Each method has advantages and disadvantages

3 Describe the three families of evaluation methods in Web search and their pros and cons.

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Offline Evaluation 101

- Offline evaluation in 3 words: Develop **test collections**
 - Collect a set of **queries**
 - For each **query**, describe the information being sought
 - Have **assessors** determine which documents are relevant
 - **Evaluate** systems based on the quality of their rankings
- **Evaluation metric**: describes the quality of ranking with known relevant/non-relevant documents

Offline evaluation has been covered at length in Lecture 5

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Offline Evaluation Cont.

Advantages

- The experimental condition is fixed; same queries, and same relevance judgements
- Evaluations are **reproducible**; keeps us “honest”
- By experimenting on the same set of queries and judgements, we can better understand how one system is better than another

Disadvantages

- Human assessors that judge documents relevant/non-relevant are **expensive**
- Human assessors are not the user; judgements are made out of context
- Assumes that relevance is the same for every user

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User Study Evaluation 101

User studies in 3 bullets:

- Provide a small set of **users** with several retrieval systems
- Ask them to complete several (potentially different) **search tasks**
- Learn about a system **performance** by
 - Observing what they do
 - Asking why they did it

The usual evaluation methods and techniques from HCI apply

D. Kelly, Methods for Evaluating Interactive Information Retrieval Systems with Users. Foundations and Trends in Information Retrieval 3(1-2): 1-224, 2009

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User Study Evaluation Cont.

Advantages

- **Detailed** data about users, and their reaction to systems
- In reality, a search is done to accomplish a higher-level **task**
- In user studies, this task can be manipulated and studied
- In other words, the experimental "starting-point" does not need to be the query

Disadvantages

- User studies are **expensive** (pay users/subjects, scientist's time, data coding, etc)
- Difficult to **generalise** from small studies to broad populations
- Environments where they are conducted are not necessarily the user's normal environment
- Need to re-run experiment every time a new system is considered

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Online Evaluation 101

- See how users **interact** with your **live** retrieval system when just using it
- Treat some of the users by a **changed version** of the system
- Based on their **behavior** (e.g. implicit feedback), infer if they are more likely to prefer the changed system
- Examples of observed **behavior**: clicks, skips, saves, forwards, bookmarks, "likes", etc.
- Two main approaches
 - **A/B testing**: Have $x/2\%$ of query traffic use system A and $x/2\%$ of query traffic use system B where x is about 5% of traffic
 - **Interleaving**: Expose a **combination** of system versions to users

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Online Evaluation

- **Assumption:** Observable user behavior reflects relevance
- This assumption gives us “**high fidelity**”
 - Real users replace the assessors: No ambiguity in information need; Users actually want results; Measures performance on real queries
- But introduces a major **challenge ...**
 - We cannot train the users: How do we know when they are happy? Real user behavior requires careful design, metrics and evaluation
- ... and noticeable **drawbacks:**
 - We need a lot of user data to compensate for **noisy** user interaction (e.g. clicks are noisy)
 - Data isn’t trivially *reusable* later

9 Describe online evaluation, its main assumption, and its pros and cons.

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Online Data & Bias

- A variety of data can describe online behavior
 - Queries, Results and **Clicks**
 - **Mouse** movement: Clicks, selections, hover
 - **Eye** tracking
- Can we simply interpret clicked results as relevant? A variety of **biases** make this difficult:
 - **Position Bias:** Users are more inclined to examine and click on higher-ranked results
 - **Contextual Bias:** Whether users click on a result depends on other nearby results
 - **Attention Bias:** Users click more on results which draw attention to themselves
 - **Accidental Clicking; Malicious Clicking, etc.**

Describe five possible biases affecting users’ interaction in Web search

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Online Evaluation

A/B TESTING

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A/B Testing (1)

- Each user is assigned to one of two conditions
- They might see the left *or* the right ranking

<http://www.profootballhof.com/>
<http://www.nfl.com/halloffame>
<http://www.stubhub.com/nfl-hall-of-fame-game-tickets/>
<http://www.mahalo.com/pro-football-hall-of-fame-game>
<http://betsportsonline.wordpress.com/2009/07/28/nfl-betting-hall-of-fame-game-odds-and-pick/>
<http://regawworld.wordpress.com/2009/08/05/betting-nfl-preseason-bills-play-titans-in-hall-of-fame-game/>



Ranking A

<http://www.profootballhof.com/>
http://en.wikipedia.org/wiki/NFL_Hall_Of_Fame_Game
<http://www.stubhub.com/nfl-hall-of-fame-game-tickets/>
<http://www.mahalo.com/pro-football-hall-of-fame-game>
http://balltype.com/story/nfl_hall_of_fame_game_2009/
<http://www.midwestsportsfans.com/2009/08/hall-of-fame-game-tickets-bills-titans-preview-odds-over-under-date-time-tv-schedule-prediction/>



Ranking B

Describe the A/B testing procedure, its pros and cons.

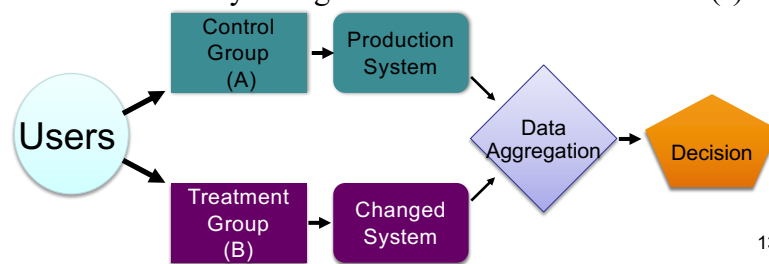
- **Measure** user interaction with their system (e.g. clicks)
- Look for differences between the populations

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A/B Testing (2)

- Concept is fairly trivial: Randomly **split traffic** between two (or more) versions
 - A (Control) & B (Treatment, i.e. Alternative System)
- Collect **metrics** of interest & Analyse
- Run **statistical tests** to confirm differences are not due to chance
- Best scientific way to demonstrate *causality* – changes in metrics are caused by changes introduced in the treatment(s)



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A/B Testing Metrics

- Examples of online **metrics** used:
 - **Abandonment Rate** (% of queries with no click)
 - Mean Reciprocal Rank (mean of 1/rank for all clicks)
 - User Engagement (e.g. clicks per Query; Time to First Click, Time to Last Click, etc)
 - Purchases (in case of ecommerce search)
 - Sessions per User
 - Probability of Switching to another search engine
- **A/B tests** are used by many web companies such as Google, Bing, Facebook, etc.
 - Use of special experimental platforms allowing to run A/B tests at **large-scale** (e.g. 100s per-day)
- A/B experiments are not the panacea for everything (c.f. survey by Kohavi et al. 2009)
 - They can take a **long time** to complete

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Advantages of A/B Testing

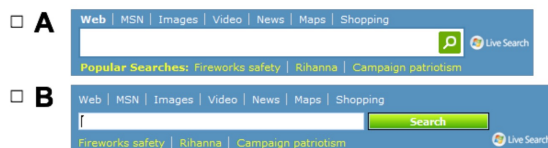
- When the variants run **concurrently**, only two things could explain a **change** in metrics:
 1. The “feature(s)” (A vs. B)
 2. Random chance
- Everything else happening affects both variants
- For #2, conduct **statistical tests** for significance (e.g. Student’s t-test)

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A/B Testing Guidelines

- Perform many **sanity checks**
 - E.g. too many unsuccessful tests means that lots of users are experiencing degraded performance
- Run an **A/A** test!
- If something is “*amazing*”, find the **flaw**
 - Look for confounding variables – e.g. ensure that not too many variables have changed



– etc.

Kohavi et al., 2013

Describe three A/B testing guidelines that help increase confidence in the results.

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Online Evaluation

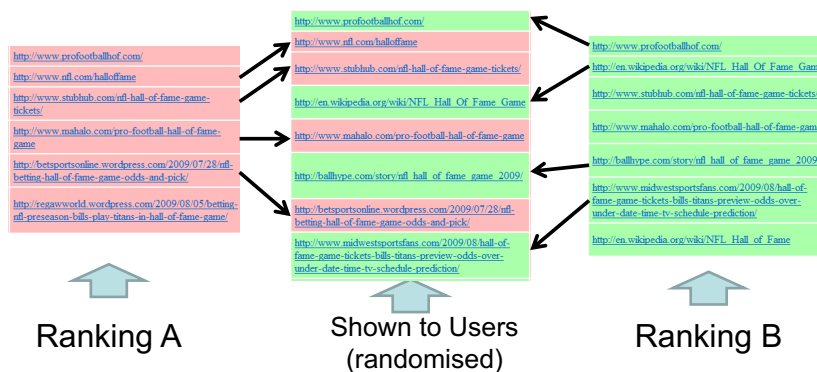
INTERLEAVING

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Online Evaluation: Interleaving

- A within-user online ranker comparison
 - Presents results from both rankings to every user



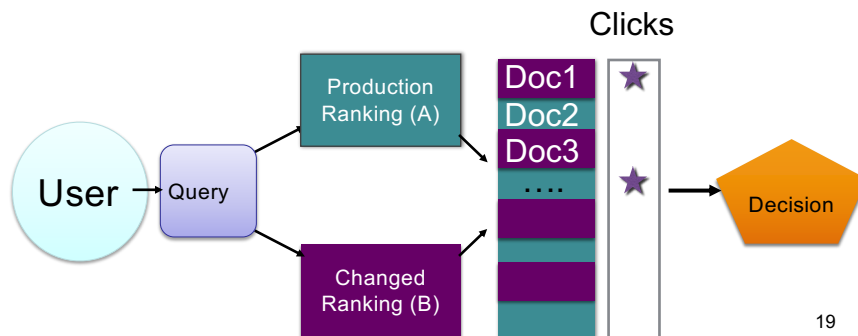
- The ranking that gets more of the clicks wins
 - Designed to be unbiased, and much more sensitive than A/B

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Interleaving

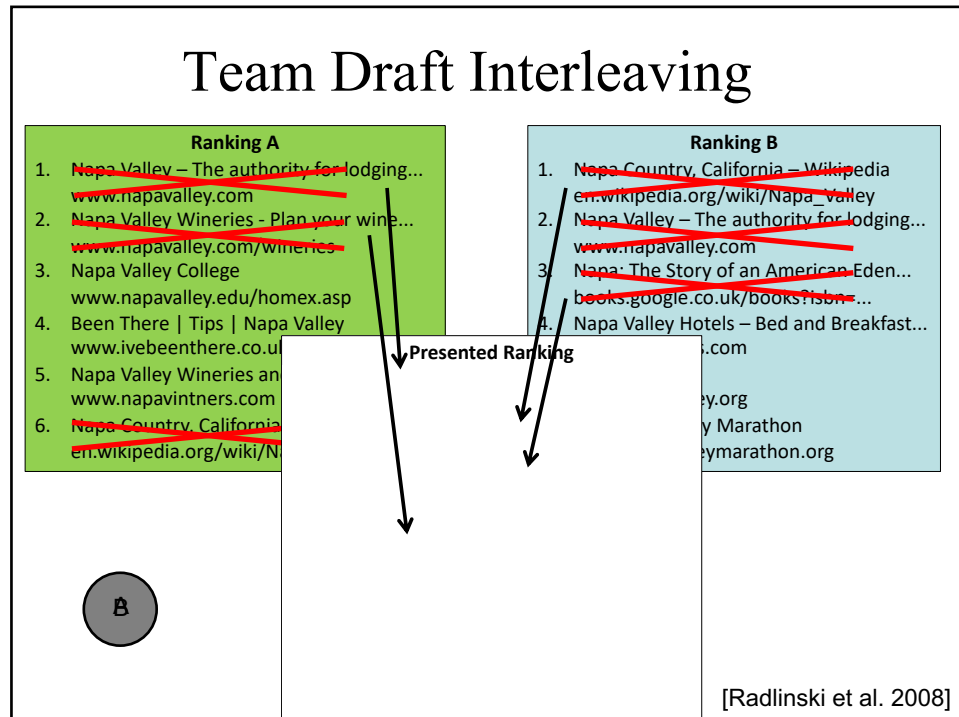
- Procedure:
 - Generate **interleaved** result list
 - Keep track of **assignments** (which ranker contributed which document)
 - Observe user **behavior** (e.g., clicks)
 - **Credit** clicks to original rankers to infer outcome



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Team Draft Interleaving



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Team Draft Interleaving

Ranking A	Ranking B	Presented Ranking
1. Napa Valley – The authority for lodging... www.napavalley.com	1. Napa Country, California – Wikipedia en.wikipedia.org/wiki/Napa_Valley	1. Napa Valley – The authority for lodging... www.napavalley.com
2. Napa Valley Wineries - Plan your wine... www.napavalley.com/wineries	2. Napa Valley – The authority for lodging... www.napavalley.com	2. Napa Country, California – Wikipedia en.wikipedia.org/wiki/Napa_Valley
3. Napa Valley College www.napavalley.edu/homex.asp	3. Napa: The Story of an American Eden... books.google.co.uk/books?isbn=...	3. Napa: The Story of an American Eden... books.google.co.uk/books?isbn=...
4. Been There Tips Napa Valley www.ivebeenthere.co.uk	4. Napa Valley Hotels – Bed and Breakfast... www.napavalley.com	4. Napa Valley Wineries – Plan your wine... www.napavalley.com/wineries
5. Napa Valley Wineries and Vineyards www.napavintners.com		5. Napa Valley Hotels – Bed and Breakfast... www.napalinks.com
6. Napa Country, California en.wikipedia.org/wiki/Napa_Valley		6. Napa Valley College www.napavalley.edu/homex.asp
		7. NapaValley.org www.napavalley.org

Click

Tie!

[Radlinski et al. 2008]

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Scoring Interleaved Evaluations

- Clicks **credited** to “owner” of result
 - Ranking r_1
 - Ranking r_2
 - Shared: A & B share top K results when they have identical results at each rank $1 \dots K$
- Ranking with more **credits** wins
- Needs a **statistical test**: e.g. Binomial test

$$\left(\mathbf{E} \left[\frac{C_A - C_B}{C} \right] > 0 \right) \rightarrow (A \succ B)$$

$$\left(\mathbf{E} \left[\frac{C_A - C_B}{C} \right] < 0 \right) \rightarrow (B \succ A)$$

C_i total clicks on results from i
 C total clicks

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Interleaving

- Examples of **metrics** used:
 - Relative difference in number of clicks received by the results from A and B
 - Ratio of the sessions with the results from B getting more clicks
- Allows to directly compare two rankings A & B. Deals with issues of **position bias** and user **calibration**.
- However, there are a number of issues:
 - **Reusability**: Can only elicit pairwise preferences for specific pairs of ranking functions
 - **Interpretation**: Doesn't tell us much about document-level assessments and user behavior

Describe the pros and cons of interleaving.

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A/B vs. Interleaving

	A/B tests	Interleaving
Idea	Treat different users with different modifications of the search engine	Treat the same user with a combination of the results from both alternatives
Applicability	Very general (UI, ranking, new products, verticals, ...)	Ranking only
Metrics used	Click-based, session-based, user-based, etc	Click-based only (somewhat restrictive)

So why do we need interleaving?

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Online Evaluation Efficiency

- It turns out that:
 - *Interleaving is more **sensitive*** = evaluating the same change using interleaving requires **10x-100x** times **less** data than the corresponding A/B test
 - It requires less data = allows us to use the resource of user sessions more **efficiently**
- Intuitive explanation:
 - In A/B tests, **different** users are treated with different systems
 - In interleaving, **the same user** compares the systems
 - ✓ The **noise** due to user variance is removed

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A/B vs Interleaving

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Metrics used	Click-based, session-based, user-based, etc	Click-based only (somewhat restrictive)
Efficiency	Not too efficient	Very efficient

What is the main advantage of using interleaving?

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Why Efficiency is Important? (1)

- «At Microsoft's *Bing*, the use of controlled experiments has **grown exponentially** over time, with over 200 concurrent experiments now running on any given day» Kohavi et al., Online Controlled Experiments at Large Scale, KDD 2013
- Running 200 experiments:
 - 10% of the **query traffic** per experiment for two weeks = 5 experiments per week = 40 weeks*
 - 5% of the **query traffic** per experiment for two weeks = 10 experiments per week = 20 weeks*

* Only a motivational example: sometimes the same user might participate in several experiments at the same time + the number of the experiments reported by Bing might span several markets.

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Why Efficiency is Important? (2)

- Number of experiments **grows**
- Each experiment consumes some **resources** (user sessions)
- The duration of the experiments limits the **evolution** of the search engine
 - The faster a change is evaluated, the faster it can be deployed

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Why Efficiency is Important? (3)

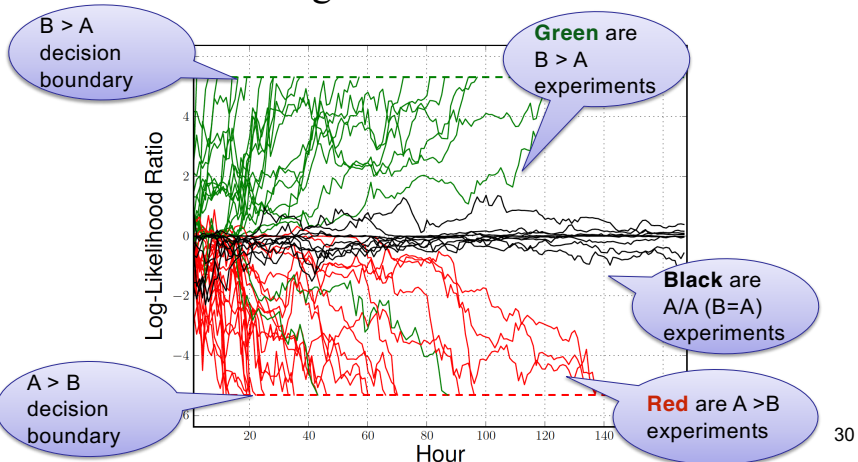
- More than a half of the tested changes are either useless or **harmful**
- On average, the users who participate in an A/B or an interleaving experiment where the tested change B is worse than the production system A, suffer a somehow **degraded experience**
- Reducing the duration of the online experiments, i.e. increasing the online evaluation **efficiency** is important since:
 - SEs do not want to harm their users' experience
 - SEs want to evolve their system as fast as possible

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Recent Research in Interleaving Efficiency (In Collaboration with Yandex, SIGIR 2015)

- **Sequential testing**: can we terminate a sequential test as soon as significance is reached?



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Online Evaluation Summary

Advantages

- System usage in a **natural environment**; users are situated in their natural context and often don't know that a test is being conducted
- Evaluation can include **a lot of users** -> better samples of the users population

Disadvantages

- Requires a service with lots of users
- Some users might experience a **sub-standard** system performance
- Requires a good understanding of how implicit feedback signals predict a +ve & -ve user experience

What are the pros and cons of online evaluation?

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